

Integral Cyclotomic Distribution Relations

Reflection torsion, modular rank loci, contact-order Smith laws,
and exact spectral-jet identities

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Abstract

The ProveIt polylogarithm manuscript proves freeness of a weighted finite cyclotomic distribution module over a characteristic-zero character splitting field and leaves its integral and modular structure for further research. We give a division-free construction over $\mathbb{Z}[A_p : p \mid q]$, including an explicit raw-point basis, a maximal relation minor equal to one, and a finite exact resolution. These tools distinguish unreflected distribution relations, which have no torsion after any integer weight specialization, from reflection coinvariants, which can acquire precisely determined 2-torsion.

For $q > 2$, put $n = \varphi(q)/2$, $r = \#\{p > 2 : p \mid q\} + \mathbf{1}_{4 \mid q}$ and $h = 2^{r-1}$. Both integer reflection signs give $\mathbb{Z}^n \oplus (\mathbb{Z}/2)^h$ when all odd-prime weights are odd, and $\mathbb{Z}^n$ otherwise. In characteristic two, the entire obstruction is a Koszul complex on the odd-prime quantities $A_p - 1$, together with $A_2(A_2 - 1)$ when $4 \mid q$. In particular its universal homology is the coordinate ring of an explicit reduced complete intersection. Along a one-parameter deformation with contact order d , the reflection matrix has h nonunit Smith factors t^d , and its length- L jet cokernel has dimension $nL + h \min(d, L)$.

Three short polynomial certificates also produce all-spectral-order cyclotomic identities and an all-derivative level-15 family involving Stieltjes constants, with the pole at order one cancelled explicitly. The ordinary distribution and its ordinary sign cohomology are classical; we identify these antecedents rather than claiming their rediscovery as new. Full proofs, exact replay programs, numerical diagnostics, a focused manuscript correction, and integration notes accompany the article.

Scope. The results concern explicitly presented modules and exact functional identities, not numerical independence of periods. No proof-assistant verification or exhaustive priority claim is made. The existing S_6 conjecture is not proved in this contribution.

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1 The research question and the distinction that matters

1.1 A precise continuation of the manuscript

The source is the ProveIt manuscript *Polylogarithms and their Arithmetic Bridges*, inspected at commit

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Its distribution-jets continuation defines independent prime weights and proves a conductor-character normal form over a characteristic-zero splitting field. Its research programme explicitly separates integral Smith forms, torsion, and modular specializations from those results [1, 2]. This article addresses that separation. The source’s positive-characteristic and integral questions are not answered by tensoring a character calculation with a smaller coefficient ring: character projectors divide by group orders, and this division is precisely what is unavailable at some primes.

The construction below uses only sums, subtraction, and multiplication by weights. Consequently it can be specialized to an arbitrary commutative ring, including a finite field or a ring with nilpotents. Reflection then becomes a second, genuinely different operation. Even when a module is free over $\mathbb{Z}$, quotienting it by $c - 1$ or $c + 1$ for an involution c can produce torsion. Rational eigenspace dimensions do not see this torsion.

The main concrete outcomes are summarized in Table 1. The proofs are ordinary algebraic and analytic proofs. Finite exact computations replay selected consequences and implementation details; they are not substitutes for the all-level arguments.

Result	Exact content and scope
Integral distribution basis	A basis of actual grid symbols over $\mathbb{Z}[A_p]$, a selected maximal minor equal to 1, and arbitrary base change.
Reflection torsion	Both signs have only 2-torsion, with an explicit weight-parity criterion and multiplicity.
Modular geometry	A Koszul description for every characteristic-two coefficient algebra; an exact field-valued rank locus.
One-parameter jets	Complete Smith exponents over $k[[t]]$, not merely a first-order rank calculation.
Cyclotomic identities	Polynomial certificates with 3, 4, and 5 raw rows; a level-15 derivative identity for every order.
Audit	A precise Clausen-parity terminology correction; integral versus field-coefficient qualifications for relation calculations.

Table 1: What this continuation proves. The classical ordinary specializations are acknowledged separately.

1.2 Classical antecedents and limits of novelty claims

The ordinary universal distribution is classical. Kubert’s 1979 papers study its generators and its $\mathbb{Z}/2\mathbb{Z}$ cohomology [3, 4]. In particular the ordinary specialization of the torsion count below recovers the known sign cohomology count; it is not a new resolution of a previously unknown ordinary case. Ouyang develops the broader universal norm distribution and an Anderson-type resolution [5]. These are important antecedents for the present finite weighted complex.

The contribution here is a self-contained, explicit treatment of the manuscript’s particular presentation, including its chosen raw-point basis, universal unit minor, arbitrary-ring specializations, full weighted reflection calculation, and contact-order Smith law. These statements supply

material absent from the inspected manuscript. We do not assert that every special case or general structural principle is absent from the wider distribution literature. The analytic identities are exact consequences of classical distribution, supplied with short certificates and all-order pole-cancelled derivatives; they are not advertised as new functional equations independent of the classical distribution law.

1.3 Notation and the formal endpoint

For a positive integer q , let

$$X_q = (q^{-1}\mathbb{Z})/\mathbb{Z}, \quad P(q) = \{p : p \text{ is prime and } p \mid q\}, \quad R_q = \mathbb{Z}[A_p : p \in P(q)].$$

We write $\text{den}(x)$ for the exact order of x in $\mathbb{Q}/\mathbb{Z}$, and set $\text{den}(0) = 1$. A symbol e_x is not itself a special value. It is a generator in a free module. Under polylogarithm evaluation, e_0 will become $\text{Li}_s(1) = \zeta(s)$. Under Hurwitz evaluation, it becomes $\zeta(s, 1)$, not the undefined expression $\zeta(s, 0)$.

For $d \mid q$, put $A_d = \prod_{p \mid d} A_p^{v_p(d)}$. Weights on composite divisors are thus multiplicative; an unrelated variable for every divisor would define a different module. The universal module and its relation submodule are

$$V_q = \bigoplus_{x \in X_q} R_q e_x, \tag{1.1}$$

$$r_{p,x} = \sum_{\substack{y \in X_q \\ py=x}} e_y - A_p e_x \quad (p \mid q, x \in X_{q/p}), \tag{1.2}$$

$$\mathcal{R}_q = \langle r_{p,x} \rangle_{R_q}, \quad \mathcal{D}_q = V_q / \mathcal{R}_q. \tag{1.3}$$

Every sum in (1.2) has exactly p terms. For a commutative ring S and weights $a_p \in S$, write $\mathcal{D}_q(S, a)$ for the same presentation after $A_p \mapsto a_p$.

Lemma 1.1 (Prime rows suffice). *The relation $r_{d,x} = \sum_{dy=x} e_y - A_d e_x$, for $d \mid q$ and $x \in X_{q/d}$, follows from the prime rows without division.*

Proof. For $ab \mid q$ and $x \in X_{q/(ab)}$, direct expansion gives

$$r_{ab,x} = \sum_{ay=x} r_{b,y} + A_b r_{a,x}.$$

Here all y belong to $X_{q/b}$, so every displayed prime or divisor row is legitimate. The intermediate symbols cancel. Induction on the number of prime factors of d , counted with multiplicity, proves the claim. $\square$

This elementary observation already prevents a common ambiguity: “distribution rank” is meaningful only after the row vocabulary, endpoint convention, coefficient ring, and divisor-weight convention have all been fixed.

2 A division-free integral basis and a unit relation minor

2.1 Choosing the surviving grid points

Write $U_m = (\mathbb{Z}/m\mathbb{Z})^\times$ and use a one-element set U_1 . For every prime p and exponent $a \geq 1$, choose a distinguished lift in each fiber of $U_{p^a} \rightarrow U_{p^{a-1}}$. At $a = 1$, this means choosing one member of U_p . Let $E_{p,a}$ be the remaining members. Thus

$$|E_{p,a}| = \varphi(p^a) - \varphi(p^{a-1}), \quad \varphi(1) = 1. \tag{2.1}$$

The following fixed choice makes the construction executable:

$$\begin{aligned} E_{p,1} &= \{2, 3, \dots, p-1\}, \\ E_{p,a} &= \{b + jp^{a-1} : 1 \leq b < p^{a-1}, p \nmid b, 1 \leq j \leq p-1\} \quad (a \geq 2). \end{aligned} \quad (2.2)$$

For $p = 2$, the first set is empty. This is intentional.

Every point x of exact denominator m has a unique additive primary expansion

$$x = \sum_{p^a \parallel m} \frac{b_p}{p^a} \pmod{\mathbb{Z}}, \quad b_p \in U_{p^a}. \quad (2.3)$$

For $x = n/m$ in lowest terms, $b_p \equiv n(m/p^a)^{-1} \pmod{p^a}$. Define $\mathcal{B}_q$ to contain 0 and all points of denominator $m \mid q$ for which $b_p \in E_{p,a}$ at every nonzero primary component. We call a primary component *bad* when its residue is the distinguished, discarded lift. This adjective refers only to a basis choice.

The cardinality telescopes prime by prime:

$$|\mathcal{B}_q| = \prod_{p^e \parallel q} \left(1 + \sum_{a=1}^e (\varphi(p^a) - \varphi(p^{a-1})) \right) = \varphi(q). \quad (2.4)$$

The point 0 must be retained for this count and for the normal forms.

Theorem 2.1 (Integral normal form). *The images of e_b , $b \in \mathcal{B}_q$, form an R_q -basis of $\mathcal{D}_q$. Consequently, for every commutative ring S and every choice of weights $a_p \in S$, the same points form an S -basis of $\mathcal{D}_q(S, a)$. In particular this module is free of rank $\varphi(q)$ even in positive characteristic, at zero weights, and over nonreduced rings.*

The spanning algorithm is immediate and is proved next. Its independence is established in Section 3, without dividing by any group order or weight.

2.2 The terminating straightening rule

Let $y \notin \mathcal{B}_q$, let $m = \text{den}(y)$, and choose any bad prime p for its primary coordinates. The row with target py gives

$$e_y = A_p e_{py} - \sum_{\substack{pz=py \\ z \neq y}} e_z \quad \text{in } \mathcal{D}_q. \quad (2.5)$$

All terms on the right have denominator dividing m . At exact denominator m , the p -primary components of the other solutions are the allowed lifts in the same fiber; all other primary components are unchanged. When the p -component has order p , there is also one solution with zero p -component, hence lower denominator. The point py itself has denominator m/p .

Order points first by their positive denominator and then by the number of bad primary components. Every right-hand term of (2.5) is strictly earlier in this order. Recursion therefore terminates, and proves that $\mathcal{B}_q$ spans. The implementation always chooses the smallest bad prime, but the theorem permits any choice. Independence will imply that all choices yield the same normal form.

Proposition 2.2 (Support and polynomial degree). *Write the normal form of e_y , with $m = \text{den}(y)$, as*

$$\text{NF}(e_y) = \sum_{b \in \mathcal{B}_q} c_{y,b}(\mathbf{A}) e_b.$$

A nonzero coefficient has $\text{den}(b) = d \mid m$, and

$$\deg_{A_p} c_{y,b} \leq v_p(m/d) \quad (p \mid q). \quad (2.6)$$

All coefficients belong to $\mathbb{Z}[\mathbf{A}]$.

Proof. The recursion never increases a primary denominator. Multiplication by A_p occurs only in the term e_{py} , where that denominator has lost one power of p . Every path from denominator m to denominator d can therefore contain at most $v_p(m/d)$ such multiplications. The other coefficients in the recursion are -1 . Summing the finitely many paths proves the assertions, including possible cancellations. $\square$

There is no assertion here that the coefficient of e_0 factors as a product of $A_p - 1$. At level 15 the coefficient in one normal form is $1 - A_3 A_5$; this distinction will matter at the spectral pole.

2.3 An explicit maximal minor equal to one

For each nonbasis point y , select the single row $r_{p,py}$ used in its first straightening step. Select the nonbasis points as columns, and order rows by their selected pivot. The restricted matrix is triangular, with every diagonal entry equal to 1: all other nonbasis columns are earlier in the terminating order.

The selected rows are distinct. Indeed a selected row has its chosen point as its unique latest nonbasis point, with coefficient 1. Two identical rows could not have different such latest points. We have proved the following stronger integrality statement.

Theorem 2.3 (Universal unit minor and relation basis). *The prime-relation matrix contains an explicitly selected $(q - \varphi(q)) \times (q - \varphi(q))$ minor equal to 1 in R_q . The selected rows form an R_q -basis of the full relation module $\mathcal{R}_q$. After specializing the weights to integers, all nonzero Smith invariants of the complete prime-relation matrix are 1; there are exactly $q - \varphi(q)$ of them.*

Proof. Triangularity proves the minor statement. Selected-row elimination reduces every vector to a combination of basis-point symbols. By Theorem 2.1, a vector in $\mathcal{R}_q$ has zero such remainder, so every raw row is a combination of the selected ones. The minor proves their independence. The same argument specializes without division. Equivalently, after an integer specialization the image lattice is saturated and has the indicated rank; the unit minor also proves directly that all Smith invariants are units. $\square$

The proof order is not circular: the resolution below proves Theorem 2.1 independently of the assertion that the selected rows span all relations.

2.4 Certificates rather than opaque elimination

The straightener can retain the exact row combination at every step. If C_y denotes a certificate for $e_y - \text{NF}(e_y)$, then

$$C_y = r_{p,py} + A_p C_{py} - \sum_{\substack{pz=py \\ z \neq y}} C_z, \quad C_b = 0 \quad (b \in \mathcal{B}_q). \quad (2.7)$$

A verifier need not trust the recursion. It can expand the listed raw rows in the original q -generator free module, multiply their explicit integer polynomials, and compare coefficients. The accompanying verifier does exactly this for the three frozen identities.

Example 2.4 (Small levels). The specified basis choices give

$$\begin{aligned}\mathcal{B}_4 &= \{0, 3/4\}, & \mathcal{B}_{12} &= \{0, 5/12, 2/3, 3/4\}, \\ \mathcal{B}_{15} &= \{0, 1/15, 4/15, 2/5, 7/15, 3/5, 2/3, 4/5\}.\end{aligned}$$

Also $\mathcal{B}_{30} = \mathcal{B}_{15}$ as sets of rational points. An exact primary component of order two has no allowed residue, which explains this last equality. It does not make the weight A_2 irrelevant to all normal-form coefficients.

The procedure terminates with polynomial coefficients, but we do not assert a polynomial-time bound in the bit length of q . There are q raw grid points to begin with, and certificate coefficients can grow. The finite support and degree bounds are rigorous; sharper complexity and coefficient-height estimates remain separate questions.

3 A finite weighted resolution and a proof of independence

3.1 The complex

Order the primes dividing q . For $S \subseteq P(q)$ put $p_S = \prod_{p \in S} p$, and define

$$C_j = \bigoplus_{\substack{S \subseteq P(q) \\ |S|=j}} R_q[X_{q/p_S}]. \quad (3.1)$$

A generator is written $[S, x]$. Define

$$d[S, x] = \sum_{p \in S} (-1)^{\#\{\ell \in S: \ell < p\}} \left(\sum_{py=x} [S \setminus \{p\}, y] - A_p[S \setminus \{p\}, x] \right). \quad (3.2)$$

The roots in each inner sum belong to $X_{q/p_{S \setminus \{p\}}}$. The augmentation $C_0 \rightarrow \mathcal{D}_q$ is the defining quotient.

Lemma 3.1. *The operators in distinct prime directions commute before the exterior signs are applied. Hence $d^2 = 0$, and the image of $C_1 \rightarrow C_0$ is exactly $\mathcal{R}_q$.*

Proof. Both successive root sums in prime directions p and ℓ enumerate once all solutions of $p\ell z = x$. The two terms consisting of one root sum and one multiplication by a weight agree in either order, and so do the terms with coefficient $A_p A_\ell$. All intermediate points lie in the required grids. The exterior signs make each pair cancel in d^2 . The assertion about C_1 is its definition. $\square$

3.2 A filtration that removes the weights

Associate to a generator the integer

$$\nu(S, x) = p_S \operatorname{den}(x), \quad (3.3)$$

which divides q . Filter the complex by increasing ν . If $p \in S$, a root of $py = x$ has denominator either $\operatorname{den}(x)$ or $p \operatorname{den}(x)$. Thus the differential does not increase ν . Its top-filtration part retains exactly the roots of denominator $p \operatorname{den}(x)$; the term with coefficient A_p lies strictly lower. The associated graded differential is consequently independent of all weights.

Fix a divisor $m \mid q$. In its associated graded summand, a generator with subset S has exact denominator m/p_S , so necessarily $S \subseteq P(m)$. For $p^a \parallel m$, the local coordinate belongs to $U_{p^{a-1}}$

when $p \in S$, and to U_{p^a} otherwise. At exponent zero the unique local coordinate is zero, represented by the one-element set U_1 .

There is a harmless CRT twist: division by a prime changes the other primary coordinates by a unit. We spell out its removal because omitting it would leave a gap in a claimed tensor-product argument. If x_ℓ is the ℓ -primary component of x , replace it in the S -summand by

$$z_\ell = \left(\prod_{p \in S \setminus \{\ell\}} p \right)^{-1} x_\ell. \quad (3.4)$$

The inverse exists on the ℓ -primary group. Under a differential edge removing p , the coordinates z_ℓ for $\ell \neq p$ are unchanged: the factor p^{-1} in a root coordinate cancels the removal of p from the product. The p -coordinate is carried to all its lifts under multiplication by p .

It follows that the associated graded summand at m is the tensor product, with its usual exterior signs, of the two-term complexes

$$R_q[U_{p^{a-1}}] \xrightarrow{N_{p,a}} R_q[U_{p^a}], \quad N_{p,a}(b) = \sum_{\substack{u \in U_{p^a} \\ u \mapsto b}} u, \quad p^a \parallel m. \quad (3.5)$$

For $a = 1$, the image of the single generator is the sum of all $p - 1$ nonzero residues. For $a \geq 2$, each fiber has p elements.

Lemma 3.2 (Local splitting). *Each $N_{p,a}$ is a split injection. Its cokernel is free with basis the images of $E_{p,a}$.*

Proof. Different fibers have disjoint supports. In one fiber, the sum of its basis vectors has a coefficient equal to 1 on the distinguished lift. Replace that one vector by the fiber sum; this is a unimodular basis change. The remaining vectors are exactly the allowed lifts. Doing this in all fibers proves both claims over any coefficient ring. $\square$

Theorem 3.3 (Exact weighted resolution). *The augmented complex*

$$0 \longrightarrow C_{|P(q)|} \longrightarrow \cdots \longrightarrow C_1 \longrightarrow C_0 \longrightarrow \mathcal{D}_q \longrightarrow 0$$

is exact. The images of $\mathcal{B}_q$ are a basis of $\mathcal{D}_q$, proving Theorem 2.1.

Proof. By Lemma 3.2, each local two-term complex is a direct sum of a contractible identity complex and its free cokernel in degree zero. Its finite tensor product is therefore acyclic in positive degrees. Its degree-zero homology has the tensor-product basis of the allowed local residues. In the graded summand at m , these are exactly the basis points of denominator m .

Now lift through the finite filtration. If a positive-degree cycle has highest nonzero filtration m , its highest component is a boundary in the associated graded complex. Subtract a lift of that boundary to obtain a cycle of lower filtration. Repetition makes the cycle a boundary in the original complex. This proves positive-degree exactness without any convergence issue.

For degree zero, suppose a linear combination of allowed points is dw with $w \in C_1$, and let M be the highest filtration present in that linear combination. If w has higher filtration, its leading component is a cycle in degree one of the associated graded complex. Positive-degree exactness there allows us to subtract a lifted boundary dv , $v \in C_2$, from w and lower its filtration without changing dw . Repetition gives a preimage of filtration at most M . Its leading boundary would now be a relation among the allowed points in the graded summand at M . Those points form a free basis of its zero-degree homology, so all their coefficients must vanish. Descending M proves independence. Spanning was proved by (2.5). Thus the claimed points form a basis of $H_0(C_\bullet) = \mathcal{D}_q$. $\square$

3.3 Base change and the rank identity

The proof works verbatim over an arbitrary commutative ring with arbitrary weights. Alternatively, the augmented complex over R_q is split exact as an underlying R_q -module complex: its degree-zero homology is free, and successive surjections onto projective kernels split. Tensoring with any R_q -algebra therefore preserves exactness. No flatness hypothesis on the target algebra is needed.

Taking the alternating ranks of the chain groups gives a useful check:

$$\text{rank } \mathcal{D}_q = \sum_{S \subseteq P(q)} (-1)^{|S|} \frac{q}{p_S} = q \prod_{p|q} (1 - p^{-1}) = \varphi(q). \quad (3.6)$$

This rank computation alone would not prove integral freeness. The local unimodular splittings and the lifted basis are the missing integral information.

Remark 3.4 (Relation to character normal forms). Over a characteristic-zero splitting field, both the conductor-character basis in the source and the present raw-point basis describe the same free module. Their transition matrix is a legitimate field-valued change of coordinates. Its construction can involve divisions by character-group orders. The raw-point basis is designed to avoid those divisions, not to replace the arithmetic convenience of character coordinates where those coordinates are available.

4 Reflection and the exact integral torsion formula

4.1 Reflection is a separate quotient operation

The map $c(e_x) = e_{-x}$ preserves the relations and satisfies $c^2 = 1$. It acts on the entire resolution by $c[S, x] = [S, -x]$. For integer weights a_p , let $D = \mathcal{D}_q(\mathbb{Z}, a)$ and define

$$D^+ = D/(c - 1)D, \quad D^- = D/(c + 1)D.$$

These are *coinvariants with a prescribed sign*, not the submodules of invariant and anti-invariant elements. Over a field of characteristic not two, the corresponding eigenspaces and coinvariants agree. Over $\mathbb{Z}$ this distinction is essential.

For any q , put

$$r(q) = \#\{p > 2 : p \mid q\} + \mathbf{1}_{4|q}. \quad (4.1)$$

For $q > 2$, we have $r(q) \geq 1$ and set $n(q) = \varphi(q)/2$, $h(q) = 2^{r(q)-1}$.

Theorem 4.1 (Weighted integral reflection theorem). *Let $q > 2$ and specialize every A_p to an integer a_p . For each $\varepsilon \in \{1, -1\}$,*

$$D/(c - \varepsilon)D \simeq \mathbb{Z}^{n(q)} \oplus (\mathbb{Z}/2\mathbb{Z})^{t(q,a)}, \quad (4.2)$$

where

$$t(q, a) = \begin{cases} h(q), & a_p \text{ is odd for every odd prime } p \mid q, \\ 0, & \text{some odd prime } p \mid q \text{ has } a_p \text{ even.} \end{cases} \quad (4.3)$$

In particular the answer is independent of the integer a_2 . At $q = 1$ or $q = 2$, the separate answers are $D^+ = \mathbb{Z}$ and $D^- = \mathbb{Z}/2\mathbb{Z}$.

The absence of odd torsion is an elementary property of an involution on a free lattice. The exact multiplicity is the substantive calculation. We prove it through the fixed points of the resolution.

4.2 Two elementary facts about involutions

For a module M with an involution c , define its two periodic cohomology groups by

$$\hat{H}^0(M) = \ker(c - 1)/(1 + c)M, \quad \hat{H}^1(M) = \ker(1 + c)/(c - 1)M, \quad (4.4)$$

and repeat with period two. These are the cohomology groups of the complex whose differentials alternate between $c - 1$ and $1 + c$, with $c - 1$ leaving even degree. This definition, rather than any unmentioned cohomology convention, fixes all signs and parity labels.

Lemma 4.2 (Torsion in sign coinvariants). *If M is a finite-rank free abelian group, every torsion class in $M/(c - \varepsilon)M$ is killed by 2. Its torsion subgroup is $\hat{H}^1(M)$ for $\varepsilon = 1$, and $\hat{H}^0(M)$ for $\varepsilon = -1$.*

Proof. Suppose $kv = (c - \varepsilon)w$ with $k \neq 0$. Applying $c + \varepsilon$ gives $k(c + \varepsilon)v = 0$, hence $(c + \varepsilon)v = 0$ because M is torsion-free. Then $(c - \varepsilon)v = -2\varepsilon v$, so the class of v is killed by 2. Conversely a vector in $\ker(c + \varepsilon)$ represents a class killed by 2. Taking its quotient by $(c - \varepsilon)M$ gives exactly the two groups in (4.4). $\square$

Lemma 4.3 (Balanced free ranks). *For every field of characteristic different from two, and arbitrary weights in that field, the two reflection coinvariants of $\mathcal{D}_q$ have dimension $\varphi(q)/2$ when $q > 2$.*

Proof. On a permutation module with u basis points and f fixed points, the two eigenspace dimensions are the integers $(u + f)/2$ and $(u - f)/2$. The sign-eigenspace functors are exact when 2 is invertible, so apply this count to the resolution. The alternating sum of the u 's is $\varphi(q)$ by (3.6). The fixed-point count for X_m is one if m is odd and two if m is even. Thus its alternating sum is

$$\sum_{S \subseteq P(q)} (-1)^{|S|} \gcd(2, q/p_S) = 0 \quad (q > 2).$$

For odd q , this is $(1 - 1)^{|P(q)|}$. If $v_2(q) = 1$, summing first over whether $2 \in S$ leaves $(2 - 1)(1 - 1)^k$, where k is the number of odd prime divisors and is positive. If $v_2(q) \geq 2$, that first sum is $2 - 2$. Hence both dimensions are $\varphi(q)/2$.

This uses integer dimension counts in the chain modules, not merely a trace equality in the coefficient field. The latter alone would not determine dimensions in positive characteristic. $\square$

4.3 Removing every nonfixed orbit from periodic cohomology

This step explains why the calculation is small despite the q original grid points. Let C_j^{mov} be the span of the nonfixed points in each summand of C_j . It is a subcomplex: if $x \neq -x$ and $py = x$, then y cannot be fixed; the term $-a_p[S \setminus \{p\}, x]$ is also nonfixed. Each term of this subcomplex is a direct sum of copies of the regular permutation module $S[\mathbb{Z}/2\mathbb{Z}]$ over the chosen coefficient ring S .

The periodic complex of a regular two-point orbit is exact over *every* commutative ring. Indeed the kernel of the difference map is the diagonal, which is the image of the sum map; the kernel of the sum map consists of $(u, -u)$, which is the image of the difference map. This assertion requires neither division by 2 nor absence of 2-torsion.

Let $K_\bullet = C_\bullet / C_\bullet^{\text{mov}}$. Its generators are only the fixed symbols $[S, 0]$ and, when available, $[S, 1/2]$. The action on $K_\bullet$ is trivial. Forming the double complex of the periodic cohomology complexes gives

$$\hat{H}^\bullet(\mathcal{D}_q(S, a)) \simeq H^\bullet \text{Tot}(T(K_\bullet)). \quad (4.5)$$

Here T denotes the alternating difference/sum complex just defined, and the chain degree of K_j contributes $-j$ to total degree.

For completeness, this comparison can be justified without assuming a spectral sequence degenerates. The augmented resolution is exact after base change, so taking horizontal homology in the double complex replaces $C_\bullet$ by D in degree zero. Removing the moving subcomplex does not change total cohomology because each of its vertical columns is exact. In both statements the horizontal width is bounded by $|P(q)|$, so in each total degree only finitely many columns occur; successive elimination along these columns justifies the two comparisons. In particular, no convergence of an infinite horizontal filtration is being used.

4.4 The fixed-point complex is a Koszul complex

For a sequence $b_1, \dots, b_r$ in a ring S , let $K_S(b_1, \dots, b_r)$ denote the tensor product of the two-term complexes $S \xrightarrow{b_i} S$, with source in chain degree one. This is the Koszul complex; its definition is all that is needed below.

For an odd prime p , the fixed-point part of its distribution map is multiplication by $1 - a_p$ on each available fixed symbol. Of the p roots of a fixed point, exactly one is fixed, and is the original point. The other roots disappear in the quotient by moving orbits. The factor at 2 has two different forms.

When $v_2(q) = 1$, its source has only 0 and its target has $0, 1/2$. Its column is

$$\begin{pmatrix} 1 - a_2 \\ 1 \end{pmatrix}. \quad (4.6)$$

The unit entry splits off an identity complex, leaving one free copy of S in degree zero. Therefore this prime contributes no Koszul generator to the fixed-point complex.

When $v_2(q) \geq 2$, both grids have two fixed points. In the order $(0, 1/2)$, the matrix is

$$B(a_2) = \begin{pmatrix} 1 - a_2 & 0 \\ 1 & -a_2 \end{pmatrix}. \quad (4.7)$$

The two roots of $1/2$ are $1/4, 3/4$, which are moving points; this explains the second column. Unimodular row and column operations transform this matrix to $\text{diag}(1, a_2(a_2 - 1))$, up to a unit sign. These operations are valid over $\mathbb{Z}[a_2]$ and over every specialization, because they pivot on the displayed 1. Odd-prime maps are scalar, so the operations commute with all other directions of the complex.

We have proved the following ring-uniform reduction.

Proposition 4.4 (Fixed-point reduction). *Over any commutative ring, $K_\bullet$ is chain-homotopy equivalent to*

$$K_S(\delta), \quad \delta = (a_p - 1 : p > 2, p \mid q) \text{ together with } a_2(a_2 - 1) \text{ if } 4 \mid q. \quad (4.8)$$

It has $r(q)$ active scalar entries. At $q = 1, 2$, this is the empty Koszul complex S in degree zero.

4.5 Completing the integer calculation

For $S = \mathbb{Z}$, the periodic complex on a trivial module alternates between 0 and multiplication by 2. It separates into adjacent two-term blocks. Tensoring one such block with the free complex $K_{\mathbb{Z}}(\delta)$ computes $K_{\mathbb{Z}}(\delta) \otimes \mathbb{F}_2$: multiplication by 2 is a free resolution of reduction modulo two. Consequently

$$\hat{H}^\varepsilon(D) \simeq \bigoplus_{\substack{0 \leq j \leq r(q) \\ j \equiv \varepsilon \pmod{2}}} H_j(K_{\mathbb{F}_2}(\bar{\delta})), \quad \varepsilon = 0, 1. \quad (4.9)$$

Only an isomorphism of abelian groups is asserted here; no extra multiplicative structure is required.

If some odd a_p is even, its active scalar reduces to 1, so its two-term complex is contractible and all the homology vanishes. Otherwise every odd-prime scalar is zero modulo two. The remaining scalar $a_2(a_2 - 1)$, when present, is also zero modulo two for every integer a_2 . All differentials in the reduced active complex therefore vanish, and

$$\dim_{\mathbb{F}_2} H_j = \binom{r(q)}{j}.$$

For $r(q) \geq 1$, the sums over even and odd j are both $2^{r(q)-1}$. Lemmas 4.2 and 4.3 now prove Theorem 4.1. When $r(q) = 0$, the only homology is in degree zero, giving the stated distinct answers for $q = 1, 2$.

4.6 Examples and the classical specialization

At level four, put $a = a_2$. The basis is $(e_0, e_{3/4})$, and

$$e_{1/2} = (a - 1)e_0, \quad e_{1/4} = a(a - 1)e_0 - e_{3/4}, \quad c = \begin{pmatrix} 1 & a(a - 1) \\ 0 & -1 \end{pmatrix}.$$

For integer a , the nonzero Smith factor of either $c - 1$ or $c + 1$ is 2, because $a(a - 1)$ is even. This two-dimensional example exhibits both the unavoidable parity and the irrelevance of the integer value of a_2 .

For all odd-prime weights odd, the following values result:

q	4	12	15	30	60	105
$n(q)$	1	2	4	4	8	24
$r(q)$	1	2	2	2	3	3
$h(q)$	1	2	2	2	4	4

For example, either sign at level 60 gives $\mathbb{Z}^8 \oplus (\mathbb{Z}/2)^4$ in the exceptional parity case, but only $\mathbb{Z}^8$ as soon as a_3 or a_5 is even.

When all $a_p = 1$, the count agrees with the classical ordinary sign-cohomology theorem of Kubert [4]; twice an odd level has the same effective prime count as that odd level. The weighted parity criterion, the fixed-point matrices, and the later jet calculation are the present explicit continuation of the manuscript's integral question.

5 Characteristic two, the universal support, and all jet lengths

5.1 A characteristic-two homology theorem over arbitrary algebras

Assume now that the coefficient ring S has characteristic two. Write $N = c - 1 = c + 1$. Then $N^2 = 0$, and both periodic cohomology groups are the same module $\ker N / \operatorname{im} N$. On a fixed-point module both periodic differentials vanish. The comparison (4.5) therefore has a particularly simple consequence.

Theorem 5.1 (Koszul model in characteristic two). *For every commutative characteristic-two ring S and every choice of its weights,*

$$\frac{\ker(N : \mathcal{D}_q(S, a) \rightarrow \mathcal{D}_q(S, a))}{\operatorname{im} N} \simeq \bigoplus_{j=0}^{r(q)} H_j(K_S(\delta)), \quad (5.1)$$

where the active sequence is (4.8).

Proof. Remove the moving-orbit subcomplex as in Section 4. On its fixed quotient the vertical periodic differential is zero. Total cohomology is therefore the direct sum of the horizontal homology groups, one from each of the finitely many chain degrees. Apply the chain-homotopy reduction of Proposition 4.4. $\square$

5.2 The universal obstruction is a reduced complete intersection

Let $R_2 = \mathbb{F}_2[A_p : p \mid q]$ and let $I_q \subseteq R_2$ be the ideal

$$I_q = (A_p - 1 : p > 2, p \mid q) + (A_2(A_2 - 1) : 4 \mid q), \quad (5.2)$$

where the second summand is omitted unless indicated. Its generators form a regular sequence. Indeed the odd-prime entries are independent linear variables shifted by one; after removing them, the possible last entry is a nonzero monic polynomial in the still-free variable A_2 . At each step multiplication by that entry is injective in the quotient by the earlier ones.

The Koszul complex on a regular sequence resolves its quotient. This can be seen inductively here without a general regular-sequence criterion: tensor a free resolution of the previous quotient with $[R_2 \xrightarrow{\delta} R_2]$. Its homology consists of the kernel and cokernel of multiplication by δ on that quotient. The kernel is zero, and the cokernel is the next quotient.

Corollary 5.2 (Universal characteristic-two support). *Over R_2 , the reflection homology is a cyclic module:*

$$\ker N / \operatorname{im} N \simeq R_2 / I_q. \quad (5.3)$$

Its zeroth Fitting ideal is exactly I_q . The closed locus defined by I_q is reduced. For odd $q > 1$ it is the single point with every weight equal to one. For $4 \mid q$ it has two points, with all odd-prime weights equal to one and $A_2 = 0$ or 1. For q twice an odd number, A_2 is free and the locus is an affine line.

Proof. The positive Koszul homology vanishes by the preceding argument; apply Theorem 5.1. A cyclic module R_2/I_q has zeroth Fitting ideal I_q by its one-generator presentation. After the odd linear equations, the only possible further equation is $A_2(A_2 - 1)$, whose two factors are relatively prime. Its quotient is a product of two fields, or of two copies of the remaining coefficient ring if one extends scalars. The stated descriptions and reducedness follow. $\square$

The cyclic universal answer does *not* imply that every fiber of the reflection homology is one-dimensional. Homology need not commute with a nonflat specialization. At an exceptional field-valued point the differentials of the specialized Koszul complex all vanish, so its homology has total dimension $2^{r(q)}$. This is the precise source of the larger fiber defect.

There is also an integral universal formulation. With the periodic convention (4.4), the same fixed-complex calculation, followed by reduction modulo two, gives

$$\widehat{H}^{2j}(\mathcal{D}_q) \simeq R_q / (2, \delta), \quad \widehat{H}^{2j+1}(\mathcal{D}_q) = 0. \quad (5.4)$$

Here $\mathcal{D}_q$ is still over the polynomial ring, not an integer weight fiber. The regularity argument is in $R_q/2R_q$. Odd periodic cohomology appearing after integer specialization is another nonflat-base-change effect, not a contradiction to (5.4).

5.3 The exact rank locus over every field

For a field k of characteristic two, a nonzero active scalar is a unit and makes the Koszul complex contractible. If all active scalars vanish, its differentials vanish and its total homology dimension is $2^{r(q)}$. To pass from homology to the desired coinvariants, note that an endomorphism N with $N^2 = 0$ on a d -dimensional vector space satisfies

$$\dim(\ker N / \operatorname{im} N) = d - 2 \operatorname{rank} N, \quad \dim \operatorname{coker} N = \frac{d + \dim(\ker N / \operatorname{im} N)}{2}. \quad (5.5)$$

Theorem 5.3 (All-field reflection dimension). *Let $q > 2$, let k be a field, and use arbitrary weights $a_p \in k$. If $\operatorname{char} k \neq 2$, both sign coinvariants have dimension $n(q)$. If $\operatorname{char} k = 2$, the signs coincide and*

$$\dim_k \mathcal{D}_q(k, a) / (c - 1) \mathcal{D}_q(k, a) = n(q) + h(q) \mathbf{1}_{\mathcal{E}_q(a)}, \quad (5.6)$$

where $\mathcal{E}_q(a)$ means

$$a_p = 1 \quad \text{for every odd } p \mid q, \quad a_2 \in \{0, 1\} \quad \text{if } 4 \mid q.$$

There is no condition on a_2 when $v_2(q) = 1$.

Proof. For characteristic different from two use Lemma 4.3. In characteristic two use the Koszul calculation and (5.5) with $d = \varphi(q)$. Over a field, $a_2(a_2 - 1) = 0$ is equivalent to $a_2 = 0$ or 1. $\square$

Over $\mathbb{F}_2$, the condition at 2 is automatic. Over $\mathbb{F}_4$ it is not: a weight satisfying $a_2^2 + a_2 + 1 = 0$ is neither zero nor one, and the reflection defect disappears when $4 \mid q$. Thus an experiment restricted to binary weights would miss part of the geometry. The exact regression suite includes all four field weights at selected levels to test this distinction.

5.4 A complete one-parameter Smith law

Let k have characteristic two, let $O = k[[t]]$, and choose power series weights $a_p(t)$. The module $\mathcal{D}_q(O, a)$ is free of rank $2n = \varphi(q)$. Form the active sequence $\delta(t)$ as before, and define its contact order by

$$d = \min_i v_t(\delta_i(t)), \quad v_t(0) = \infty. \quad (5.7)$$

The word “contact” refers to the pullback of the explicit ideal I_q along the chosen formal curve. It is not an approximation to a root of a transcendental function.

Each active prime contributes a factor at least two to $\varphi(q)$, so $\varphi(q) \geq 2^{r(q)}$ and $n \geq h$. Thus all block sizes in the following statement are nonnegative.

Theorem 5.4 (Contact-order Smith law). *Let $q > 2$, $n = n(q)$, $h = h(q)$, and use the above characteristic-two power series weights. If $d < \infty$, the Smith diagonal of $N = c - 1$ over O is, up to units,*

$$\operatorname{diag}(\underbrace{1, \dots, 1}_{n-h}, \underbrace{t^d, \dots, t^d}_h, \underbrace{0, \dots, 0}_n). \quad (5.8)$$

When $d = 0$, the first two blocks together are n unit entries. When $d = \infty$, replace the h entries t^d by zero; there are then $n - h$ unit entries and $n + h$ zeros.

For every integer $L \geq 1$, put $O_L = k[t]/(t^L)$. The complete jet cokernel formula is

$$\dim_k \frac{\mathcal{D}_q(O_L, a)}{(c - 1) \mathcal{D}_q(O_L, a)} = nL + h \min(d, L), \quad (5.9)$$

with $\min(\infty, L) = L$.

Proof. First work over O_L and put $d_L = \min(d, L)$. Elementary invertible operations on the active sequence identify its Koszul complex with that on $(t^{d_L}, 0, \dots, 0)$, with the convention that $t^L = 0$. To see this explicitly when $d_L < L$, choose an entry with least valuation, multiply it by a unit to make it t^{d_L} , and subtract its multiples from all other entries. Every other entry is divisible by it in O_L . If $d_L = L$, all entries already vanish. Invertible changes of the sequence induce invertible changes on its exterior algebra, hence isomorphisms of Koszul complexes.

The kernel and cokernel of multiplication by t^{d_L} on O_L each have k -dimension d_L . Tensoring with $r - 1$ zero-differential two-term complexes multiplies total homology dimension by 2^{r-1} . The right-hand side of (5.1) therefore has k -dimension $2^r d_L$. The underlying vector space of $\mathcal{D}_q(O_L, a)$ has dimension $2nL$, by the integral basis theorem. Applying (5.5) proves (5.9).

It remains to show that these length formulas determine all Smith exponents, rather than just their sum. A matrix over the discrete valuation ring O has nonzero invariant factors t^{e_i} up to units, together with some zero factors. If there are z zero factors, its cokernel modulo t^L has length

$$zL + \sum_i \min(e_i, L).$$

The first difference in L is $z + \#\{i : e_i \geq L\}$ for $L \geq 1$. For finite d , the formula already proved has first difference $n + h$ for $1 \leq L \leq d$, and n for $L > d$. Thus exactly n factors are zero and exactly h nonunit factors have exponent d ; the remaining $n - h$ factors are units. For $d = 0$ there are no positive exponents. For $d = \infty$, the first difference is always $n + h$, so there are $n + h$ zero factors and no positive finite exponents. This proves every case of (5.8). $\square$

5.5 Examples of the jet law

At level 15, take

$$a_3(t) = 1 + t^2, \quad a_5(t) = 1 + t^5.$$

Here $(n, r, h) = (4, 2, 2)$ and $d = 2$. The reflection matrix has Smith entries $1, 1, t^2, t^2, 0, 0, 0, 0$, and its jet cokernel dimensions at $L = 1, 2, 3, 4$ are 6, 12, 16, 20. A rank calculation at $t = 0$ alone would detect the first number, but not the exponent two.

At level 12, take $a_2(t) = t^3$ and $a_3(t) = 1 + t^5$. Then $a_2(a_2 - 1) = t^3(1 + t^3)$, so $d = 3$. Since $(n, h) = (2, 2)$, there are no unit Smith factors and the diagonal is $t^3, t^3, 0, 0$. Every jet of length at most three has trivial reflection matrix up to equivalence; later lengths detect the nonzero factors.

If the entire curve is contained in the exceptional locus, then $d = \infty$ and the dimension is $(n + h)L$. If any active scalar has a nonzero constant term, $d = 0$ and the dimension is nL . These are exact boundary cases of the same formula. In contrast, the *unreflected* distribution module always has dimension $\varphi(q)L$ at length L and never exhibits this jump.

6 Exact cyclotomic identities and pole-cancelled spectral jets

6.1 Evaluating an integral certificate

For $x \in \mathbb{Q}/\mathbb{Z}$, write

$$\mathcal{L}_s(x) = \text{Li}_s(e^{2\pi i x}).$$

At $x = 0$, this means $\zeta(s)$. For $x \neq 0$, the spectral function is entire. One way to see this, and to fix its continuation, is the finite residue-class formula

$$\mathcal{L}_s(a/q) = q^{-s} \sum_{u=1}^q e^{2\pi i a u/q} \zeta(s, u/q). \quad (6.1)$$

For $a \not\equiv 0 \pmod{q}$, the residues of the simple poles at $s = 1$ sum to zero. The standard meromorphic continuation of Hurwitz zeta has no other spectral poles [6].

For $\Re s > 1$, absolute convergence gives

$$\sum_{py=x} \mathcal{L}_s(y) = p^{1-s} \mathcal{L}_s(x). \quad (6.2)$$

Indeed the root sum cancels every series index not divisible by p ; for index pk it contributes $pe^{2\pi i kx}$. Continuation then proves the meromorphic identity for every complex s . Thus an integral polynomial certificate becomes an exact analytic identity under

$$e_x \mapsto \mathcal{L}_s(x), \quad A_p \mapsto p^{1-s}, \quad (6.3)$$

where real positive p fixes the exponential unambiguously. No integer relation search or floating-point acceptance threshold occurs in this step.

Proposition 6.1 (A removable endpoint coefficient). *For $y \neq 0$, the coefficient $c_{y,0}(\mathbf{A})$ in its integral normal form vanishes at $A_p = 1$ for all p . Hence its polylogarithmic specialization cancels the simple pole of $\zeta(s)$ at $s = 1$.*

Proof. At ordinary weights, the functional taking e_0 to 1 and all other grid symbols to 0 annihilates every distribution row. For $x = 0$, both the root sum and e_x contain exactly one copy of e_0 ; for $x \neq 0$, neither does. Apply the functional to the normal form of e_y . The second assertion follows because $p^{1-s} = 1$ at $s = 1$ and the coefficient is analytic in s . $\square$

The vanishing need only be first order. It is not legitimate to infer one vanishing factor for each prime merely from the number of prime variables in a raw-point normal form.

6.2 A three-row level-twelve identity

In the level-12 free module, define

$$F_{12} = e_{1/12} + e_{5/12} + (A_3 + 1)e_{3/4} - A_2A_3(A_2 - 1)e_0.$$

The following is an equality before taking the quotient:

$$F_{12} = r_{3,1/4} + A_3r_{2,1/2} + A_2A_3r_{2,0}. \quad (6.4)$$

For example, the first row contains $e_{1/12} + e_{5/12} + e_{3/4} - A_3e_{1/4}$; the next two rows cancel $e_{1/4}$ and then $e_{1/2}$. This verifies the certificate by direct inspection.

Corollary 6.2. *For every complex s , with the removable value used at $s = 1$,*

$$\mathcal{L}_s(1/12) + \mathcal{L}_s(5/12) + (3^{1-s} + 1)\mathcal{L}_s(3/4) = 6^{1-s}(2^{1-s} - 1)\zeta(s). \quad (6.5)$$

At $s = 1$, the right-hand side is $-\log 2$.

Proof. Evaluate (6.4) using (6.3). The value at one follows from $2^{1-s} - 1 = -(s - 1)\log 2 + O((s - 1)^2)$. $\square$

6.3 A four-row level-fifteen identity

Set

$$U = \{1/15, 4/15, 7/15, 2/5, 4/5\}. \quad (6.6)$$

In the level-15 free module define

$$F_{15} = e_{8/15} - \sum_{u \in U} e_u - (A_3 + 1)e_{3/5} - (A_5 + 1)e_{2/3} - (1 - A_3A_5)e_0. \quad (6.7)$$

A short exact certificate is

$$F_{15} = r_{3,3/5} - r_{5,1/3} - A_5 r_{3,0} - r_{5,0}. \quad (6.8)$$

The roots in the first two rows are respectively

$$\{1/5, 8/15, 13/15\}, \quad \{1/15, 4/15, 7/15, 2/3, 13/15\}.$$

The $13/15$ terms cancel. The last two rows remove $1/3$ and $1/5$, and leave precisely (6.7). In particular the coefficient of e_0 in the normal form of $e_{8/15}$ is $1 - A_3A_5$, not $(A_3 - 1)(A_5 - 1)$.

Theorem 6.3 (All-order level-fifteen identity). *For every complex spectral order s ,*

$$\begin{aligned} \mathcal{L}_s(8/15) - \sum_{u \in U} \mathcal{L}_s(u) - (3^{1-s} + 1)\mathcal{L}_s(3/5) - (5^{1-s} + 1)\mathcal{L}_s(2/3) \\ = (1 - 15^{1-s})\zeta(s). \end{aligned} \quad (6.9)$$

Both sides extend holomorphically through $s = 1$, where the right-hand side is $\log 15$.

Proof. Apply (6.3) to the four-row identity. The endpoint value follows from $1 - 15^{1-s} = (s - 1) \log 15 + O((s - 1)^2)$ and the residue one of $\zeta(s)$. $\square$

The theorem supplies infinitely many exact identities, including ordinary positive integer weights, negative orders, and derivatives in spectral order. It is generated by classical distribution rather than by a new numerical relation conjecture.

A compatible level-30 normal form follows from the extra row $r_{2,1/15}$:

$$\begin{aligned} e_{1/30} = (A_2 - 1)e_{1/15} - (A_3 + 1)e_{3/5} - (A_5 + 1)e_{2/3} \\ - e_{4/15} - e_{7/15} - e_{2/5} - e_{4/5} + (A_3A_5 - 1)e_0. \end{aligned} \quad (6.10)$$

Its residual has the five-row certificate

$$r_{2,1/15} - r_{3,3/5} + r_{5,1/3} + A_5 r_{3,0} + r_{5,0}.$$

All rows can be read in the level-30 grid. This illustrates compatibility of the straightener with an enlarged level and a weight whose order-one specialization vanishes in a particular coefficient.

6.4 Every spectral derivative, with the pole term retained

Let

$$\mathcal{L}^{[j]}(x) = \left. \frac{\partial^j}{\partial s^j} \mathcal{L}_s(x) \right|_{s=1}, \quad x \neq 0,$$

and write the Laurent expansion

$$\zeta(1+t) = \frac{1}{t} + \sum_{m \geq 0} \frac{(-1)^m \gamma_m}{m!} t^m. \quad (6.11)$$

For a positive prime p and $N \geq 0$, define the explicit finite operator

$$\mathcal{T}_{p,N}(x) = \mathcal{L}^{[N]}(x) + \sum_{j=0}^N \binom{N}{j} (-\log p)^j \mathcal{L}^{[N-j]}(x). \quad (6.12)$$

It is the N th derivative of $(p^{1-s} + 1)\mathcal{L}_s(x)$ at $s = 1$.

Theorem 6.4 (A level-fifteen Stieltjes–polylogarithm jet family). *For every integer $N \geq 0$, put $\ell = \log 15$. Then*

$$\begin{aligned} & \mathcal{L}^{[N]}(8/15) - \sum_{u \in U} \mathcal{L}^{[N]}(u) - \mathcal{T}_{3,N}(3/5) - \mathcal{T}_{5,N}(2/3) \\ &= (-1)^N \left\{ \frac{\ell^{N+1}}{N+1} - \sum_{j=1}^N \binom{N}{j} \ell^j \gamma_{N-j} \right\}. \end{aligned} \quad (6.13)$$

The sum on the right is empty at $N = 0$.

Proof. All nonzero-root terms of (6.9) are entire in s , so differentiate the holomorphic continuation N times. The weighted terms give (6.12) by Leibniz’s rule. For the right side use

$$1 - e^{-\ell t} = \sum_{j \geq 1} \frac{(-1)^{j+1} \ell^j}{j!} t^j$$

and multiply by (6.11). The pole term contributes $(-1)^N \ell^{N+1}/(N+1)$ to the N th derivative. The regular Laurent coefficients contribute $(-1)^{N+1} \sum_{j=1}^N \binom{N}{j} \ell^j \gamma_{N-j}$. This is the stated formula. $\square$

The first derivative gives the especially compact identity

$$\begin{aligned} & \mathcal{L}^{[1]}(8/15) - \sum_{u \in U} \mathcal{L}^{[1]}(u) - 2\mathcal{L}^{[1]}(3/5) - 2\mathcal{L}^{[1]}(2/3) + \log 3 \mathcal{L}^{[0]}(3/5) + \log 5 \mathcal{L}^{[0]}(2/3) \\ &= \gamma_0 \log 15 - \frac{1}{2} \log^2 15. \end{aligned} \quad (6.14)$$

For $N = 2$, the scalar right-hand side is

$$\frac{\log^3 15}{3} - \gamma_0 \log^2 15 - 2\gamma_1 \log 15.$$

These terms would be wrong if one substituted $s = 1$ into the coefficient of $\zeta(s)$ before cancelling its pole.

More generally, if an analytic coefficient $c(t)$ satisfies $c(0) = 0$, then

$$\left. \frac{d^N}{dt^N} \{c(t)\zeta(1+t)\} \right|_{t=0} = \frac{c^{(N+1)}(0)}{N+1} + \sum_{j=0}^N \binom{N}{j} c^{(j)}(0) (-1)^{N-j} \gamma_{N-j}. \quad (6.15)$$

This coefficient identity gives a general implementation rule for every raw-point certificate, not just the displayed level.

6.5 An elementary endpoint and branch check

For $0 < x < 1$, the chosen branch gives

$$\mathcal{L}^{[0]}(x) = -\log(2 \sin \pi x) + i\pi(1/2 - x).$$

Taking real parts of (6.9) at $s = 1$ gives

$$\frac{\prod_{u \in U} (2 \sin \pi u) (2 \sin(3\pi/5))^2 (2 \sin(2\pi/3))^2}{2 \sin(8\pi/15)} = 15.$$

Since $\sin(8\pi/15) = \sin(7\pi/15)$, it simplifies to

$$(2 \sin(\pi/15))(2 \sin(4\pi/15))(2 \sin(2\pi/5))^3 (2 \sin(\pi/5)) = 5. \quad (6.16)$$

All factors are positive, so exponentiating the real logarithmic identity introduces no ambiguity. The imaginary parts cancel by the displayed linear argument formula. This is an elementary specialization and branch check, not a claim of novelty for the trigonometric product itself.

6.6 A division-free Hurwitz–Stieltjes translation

The same polynomial certificates evaluate under

$$A_p = p^s, \quad e_x \mapsto Z_x(s), \quad Z_x(s) = \begin{cases} \zeta(s, x), & 0 < x < 1, \\ \zeta(s, 1), & x = 0. \end{cases}$$

The distribution identity here is $\sum_{py=x} Z_y(s) = p^s Z_x(s)$; it follows by splitting the Hurwitz series into residue classes, including the endpoint convention. Let $c_b(t)$ be the normal-form coefficient of e_b for e_x after $A_p = p^{1+t}$, and put $C(t) = \sum_b c_b(t)$. Residue comparison gives $C(0) = 1$. If $\gamma_j(b^*)$ denotes the generalized Stieltjes coefficient with $b^* = b$ for $b \neq 0$ and $b^* = 1$ for $b = 0$, expansion of the exact normal form gives, for all $N \geq 0$,

$$\gamma_N(x^*) = \frac{(-1)^N C^{(N+1)}(0)}{N+1} + \sum_{b \in \mathcal{B}_q} \sum_{j=0}^N (-1)^j \binom{N}{j} c_b^{(j)}(0) \gamma_{N-j}(b^*). \quad (6.17)$$

To verify the sign, multiply each Laurent series by $c_b(t)$, compare t^N , and multiply by $(-1)^N N!$. The pole part is $C(t)/t$ and contributes the first term. Thus even the underived constants require a coefficient derivative. This is the raw-basis version of the manuscript's pole-corrected jet philosophy, now with division-free integral certificates for the coefficients.

Every coefficient derivative in (6.17) is a finite integer combination of products of logarithms of primes and integer prime powers. There is no need to differentiate a solved matrix inverse or to redo a relation search separately at each derivative order.

7 Reproducible verification and what the computations establish

7.1 The exact core and its independent certificate replay

The package's core is ordinary Python integer arithmetic; it has no third-party dependency. A sparse polynomial is encoded as integer coefficients indexed by exponent tuples in the ordered prime variables. No source string is passed to an evaluator, and no floating-point number enters a polynomial equality check.

The file `distribution.py` constructs the basis, terminates the straightener, and produces raw-row certificates. The independent `verify_certificates.py` does not call the basis selector, straightener, or certificate generator. It reads the frozen JSON, reconstructs each prime fiber directly in the original q -point grid, expands the polynomial row combination, and checks exact coefficient cancellation. It shares only the sparse integer-polynomial arithmetic. A deliberate one-coefficient corruption is also tested and rejected. This controls a simple failure mode of the replay code; it is not a transcendental-independence test.

The three certified identities use 3, 4, and 5 raw rows at levels 12, 15, and 30, respectively. Their selected universal unit minors have sizes 8, 7, and 22. The coefficients and row indices are preserved in `data/identity_certificates.json`; the exact replay result is in `logs/certificate_replay.json`.

7.2 Exact regression coverage

The fresh exact tests produced the counts in Table 2. They check a range of specializations, including ones where character diagonalization would be unavailable. They do not establish the all-level theorems by finite extrapolation; those rest on the proofs above.

Exact check	Count
Universal levels: $1 \leq q \leq 120$, plus 150, 180, 210	123
Raw prime rows reducing to zero as integer polynomials	3905
Triangular universal unit-minor checks	123
Normal-form support and multidegree checks	25628
Binary weight specializations of the reflection rank	478
Weight specializations over $\mathbb{F}_4$	240
Characteristic-two truncated-series jet cases	385
Resolution checks over $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5$	123
Integer Smith forms for both reflection signs	576
Independent frozen identity-certificate replays	3

Table 2: Fresh exact computations included with this delivery.

The resolution tests construct its matrices from (3.2), verify $d^2 = 0$, and check that adjacent boundary ranks add to the chain dimension in every positive degree. They use $2 \leq q \leq 40$ and $q = 60, 105$, in each of the three specified characteristics. Their degree-zero quotient dimension is checked against $\varphi(q)$.

The jet tests work with bit-encoded polynomials in $\mathbb{F}_2[t]/(t^L)$, for $1 \leq L \leq 7$, eleven levels, and five weight profiles. They expand the reflection operator as a binary matrix on all basis-point/jet-degree pairs and compute its exact rank. The profiles include a curve entirely on the exceptional locus, different positive contact orders, the $a_2 = 0$ branch, and a generic point. The predicted contact order is computed directly from the active scalars, not fitted to the matrix rank.

The optional integer Smith tests use SymPy’s integer Smith algorithm, independently of the binary-rank formula. At fifteen specified levels, they use all combinations of weights in $\{-1, 0, 1, 2\}$ and both reflection signs. They verify $c^2 = 1$ as an integer matrix and compare every invariant factor, including the zeros. Exact logs record the versions used: Python 3.13.5 and SymPy 1.14.0. These version strings report the executed environment, not requirements for a future run.

7.3 Numerical diagnostics use a separate analytic representation

The analytic equalities already have exact proofs. Numerical work is included to expose convention and branch errors, not to upgrade a small residual into a proof. The level-15 spectral identity was checked at $s = 2, 3, 1/2, 1.2 + 0.4i, -2, 1$, at both 65 and 95 decimal working digits. At 95 digits, the largest recorded absolute residual in these six checks is approximately 1.82×10^{-94} .

For an independent mathematical representation, use

$$\text{Li}_s(z) = \frac{1}{\Gamma(s)} \int_0^\infty u^{s-1} \frac{ze^{-u}}{1 - ze^{-u}} du, \quad \Re s > 0, \quad |z| = 1, \quad z \neq 1. \quad (7.1)$$

This is the standard Mellin representation [7]; it also follows by first working with $|z| < 1$ and passing to the boundary away from $z = 1$. Its integral is locally uniformly convergent in $\Re s > 0$. Near zero, the kernel is bounded, and $u^{\sigma-1} |\log u|^N$ is integrable for $\sigma > 0$; at infinity there is exponential decay. These bounds justify spectral differentiation under the integral on compact subsets.

Instead of numerically differencing polylogarithm order at an integer, we differentiate (7.1) analytically. The first four polynomial multipliers are

$$\begin{aligned} P_0(X) &= 1, \\ P_1(X) &= X + \gamma_0, \\ P_2(X) &= (X + \gamma_0)^2 - \zeta(2), \\ P_3(X) &= (X + \gamma_0)^3 - 3\zeta(2)(X + \gamma_0) + 2\zeta(3). \end{aligned}$$

They are the derivatives of $e^{tX}/\Gamma(1+t)$ at zero. A factor p^{-t} replaces X by $X - \log p$. Integrating the corresponding combined level-15 kernels tests (6.13) for $0 \leq N \leq 3$. At 65 working digits, the largest recorded residual among these four checks is approximately 5.95×10^{-63} . Separate Mellin checks at $s = 1, 2$ also agree with the direct polylogarithm evaluation.

These are floating-point quadrature diagnostics. No interval enclosure or rigorous numerical error bound is claimed. A recorded zero residual means zero at the working precision, not a newly certified exact zero. All 18 diagnostics, their precisions, and their residuals are preserved in `logs/numerical_diagnostics.json`. The exact equality certificates remain the polynomial identities and the analytic argument translating them.

7.4 Replay commands

From the package root, the exact core is replayed by

```
python code/verify_certificates.py
python code/test_exact.py
```

The optional independent Smith and numerical diagnostics are

```
python code/test_smith.py
python code/numerical_diagnostics.py
```

The first optional command requires SymPy; the second requires mpmath. The delivered versions are pinned in an optional requirements file. The certificate generator is provided separately, so replay does not silently regenerate a claimed certificate before checking it.

A Makefile builds the article with three LaTeX passes and exposes the same checks. The supplied PDF has been compiled and visually inspected. Neither these programs nor the prose have been checked in Lean or another proof assistant. The theorem statements and algebraic interfaces are suitable future formalization targets, not already formalized results.

8 Manuscript audit, integration, and further research

8.1 A concrete Clausen-parity correction

The inspected chapter `chapters/02-cyclotomic.tex` explicitly adopts

$$\mathrm{Cl}_{2m}(\theta) = \Im \mathrm{Li}_{2m}(e^{i\theta}), \quad \mathrm{Cl}_{2m+1}(\theta) = \Re \mathrm{Li}_{2m+1}(e^{i\theta}).$$

Here Cl denotes the manuscript’s Clausen notation. Later, under “The fundamental constants, by conductor”, the sentence “Odd Clausen values use the odd character sector” is inconsistent with this weight-index convention [1]. Odd-index Clausen values are cosine-type real parts and use even characters, as the chapter’s own $\mathrm{Cl}_3(2\pi/5)$ formula with the even quadratic character already illustrates.

A precise replacement is:

Imaginary, sine-type cyclotomic values use the odd character sector; real, cosine-type values use the even sector. Under the Clausen convention used here, even-index Clausen functions belong to the odd sector and odd-index Clausen functions belong to the even sector.

This is a terminology correction at the identified occurrence. It does not invalidate the correctly stated Fourier formulas elsewhere in that chapter. The integration directory supplies a guarded replacement script and the exact old and new text; it does not modify the repository during this delivery.

8.2 An integral qualification, not a refutation of field results

The manuscript’s descriptions of Fourier-conjugate relations should always retain their coefficient field. A Fourier transform is invertible over a suitable cyclotomic field. It is not automatically a unimodular change of integral lattices, since its inverse has a denominator. Accordingly, agreement of field-valued ranks does not prove agreement of integral Smith forms. This is a clarification of scope, not a claim that the manuscript’s explicitly field-valued identities are false.

The present calculations show why the qualification matters. Before reflection, the chosen integral presentation has a unit maximal minor and no torsion. After reflection, the quotient can contain a nonzero $(\mathbb{Z}/2)^h$, although both characteristic-zero sign dimensions are still $\varphi(q)/2$. In characteristic two, the rank jump is controlled by the active ideal (5.2), not by a failure of freeness of the unreflected distribution module.

It would therefore be inaccurate to summarize the result as “distribution rank jumps at exceptional weights”. The correct statement is that the *reflection coinvariant* dimension can jump in characteristic two. The original distribution module remains free with the same raw-point basis at those weights.

8.3 How to integrate the results without changing older claims

A natural insertion is a section following the existing distribution normal-form discussion, with cross-references from the surviving research programme. The complete statement should include Theorems 2.1, 4.1, 5.1, and 5.4. The fixed-point reduction is short enough to retain rather than hide behind an unexplained rank calculation. The analytic application can be inserted beside the existing spectral-jet identities, with (6.8) retained as a compact executable certificate.

The original characteristic-zero conductor basis remains useful and correct within its stated scope. Its complex spectral weights should not be reduced “modulo two” without first choosing

a genuine integral or modular coefficient model. The formal characteristic-two jets in this article are power-series deformations of algebraic weights; they are not reductions of real logarithms or complex spectral parameters. Both uses of the word “jet” are explicit here, but their coefficient categories are different.

The ordinary specialization is to be cited to the classical distribution literature. The new integration should not relabel it as a first proof. The existing mixed Gaussian S_6 conjecture retains its current status: none of the depth-one module calculations in this article proves that depth-two period identity. Similarly, no new numerical independence claim follows from any formal free basis or nonzero module obstruction.

8.4 Further research questions

Integral equivariant structure beyond reflection. The proof treats one involution. For a subgroup of $(\mathbb{Z}/q\mathbb{Z})^\times$ of higher order, determine the integral coinvariants and their primary torsion. An appropriate orbit filtration would have to distinguish several stabilizer types, rather than only fixed and moving points. A first concrete target is a subgroup of prime order and a coefficient characteristic equal to that prime. The raw basis and the finite resolution give an explicit starting presentation.

A full equivariant normal form. The integral basis is deliberately not reflection-stable. Can one give an explicit decomposition of the specialized lattice into trivial, sign, and regular involution blocks, together with unimodular change matrices of controlled size? The torsion calculation determines strong constraints on such a decomposition, but the short fixed-complex proof does not itself output those basis changes.

Multivariate deformation invariants. Theorem 5.1 identifies every characteristic-two base change through Koszul homology. Over a multivariate power-series ring there need not be a Smith diagonal. Determine a useful replacement in terms of Fitting ideals, minimal free resolutions, and intersection multiplicities of the active ideal. The complete-intersection support and the one-variable contact law provide tests for any proposed multivariate formula. In particular, the universal cyclic homology must not be naively specialized as if homology were flat.

Explicit transition to conductor-character coordinates. Construct the matrix between the raw-point basis and the manuscript’s conductor basis over cyclotomic integers localized at specified primes. Determine its denominator ideal, not merely its determinant over a splitting field. This would isolate exactly which arithmetic primes obstruct character-coordinate replay and permit certified transport between Fourier and raw-row certificates.

Shortest certificates and coefficient growth. The selected-row straightener gives one certificate, not necessarily a shortest one. Determine coefficient-height and sparsity bounds as functions of q , its prime factorization, and the target denominator. At levels with many prime factors, compare different distinguished-lift choices and pivot orders. A concrete optimization problem is to minimize the number of raw rows for a fixed normal-form identity while keeping integer polynomial coefficients.

Higher-depth distribution. At depth one, every root sum has a single target and the resolution separates prime directions. Multiple polylogarithms introduce nested index constraints, several colors, and shuffle/stuffle compatibility. Define an integral higher-depth presentation before asking

for its rank or torsion. It would be especially useful to identify which parts of the present unit-pivot argument survive and which introduce new syzygies. This is a route toward stronger exact certificate infrastructure, not a present proof of the S_6 conjecture.

Pole-aware identity generation. Use the degree bound (2.6) to generate families of spectral derivatives at nonpositive integers and at the simple pole, while keeping the endpoint contribution symbolic. Seek selections whose real or imaginary parts remove most derivative coordinates. The all-order family (6.13) is a small test case; arithmetic reduction of its surviving period coordinates remains a separate problem.

Certified numerics and formal proof. Replace the floating-point Mellin diagnostics by interval quadrature with explicit endpoint and tail budgets. Independently, formalize finite grids, the straightening termination order, and raw certificate replay in Lean. The next algebraic layer is the split local complex and finite-filtration proof; the periodic fixed-point reduction can then be treated as a separate homological interface. These tasks would convert precisely identified pieces of the present ordinary proofs into machine-checked results.

8.5 Conclusion

The continuation resolves the integral and modular structure of the specified depth-one weighted distribution presentation, while keeping its analytic meaning and its limitations explicit. A basis can remain free under every specialization even though a later symmetry quotient acquires torsion. The fixed-point Koszul complex explains both the integer parity rule and the full characteristic-two jet law. At the analytic end, short polynomial row certificates give exact identities at every spectral order, and their endpoint derivatives force a visible Stieltjes correction. These are complementary uses of the same integral presentation, rather than unrelated numerical discoveries.

References

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